

The Kilim plot: a tool for visualizing network meta-analysis results for multiple outcomes

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Background

- **Network meta-analysis** (NMA) can be used to compare multiple treatments for the same disease. A range of outcomes is usually of interest.
- It is difficult to efficiently summarize results from NMAs on **multiple outcomes**, especially when the number of treatments and/or outcomes is large.
- NMAs often provide results in terms of **relative effect measures** that can be difficult to apply in **every-day clinical practice**, such as the odds ratios.

Aims

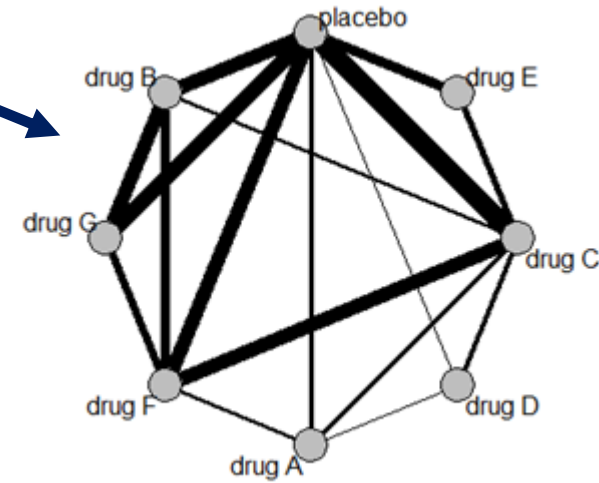
- Our predefined goals when developing the **Kilim plot** were the following:
 - (1) *to provide a method for visualizing the evidence from **multiple outcomes NMAs**.*
 - (2) *to present results in terms of **absolute**, rather than relative effects.*
 - (3) *to illustrate graphically the evidence with respect to **clinically important values**.*

Clinical example: side effects of antidepressants

- We used 297 RCTs on **antidepressants** for the **acute treatment of depression**.
- The studies compared **seven antidepressants and placebo** with respect to **nine different side effects** (all binary outcomes).
- Drug names have been **anonymized**.
- Three of the outcomes were **serious side effects**, i.e. *suicidal ideation, aggression, and accidental overdose*.
- Six outcomes corresponded to **frequent side effects**, i.e. *nausea, headache, dry mouth, insomnia, sexual dysfunction, and diarrhoea*.

We would have **nine** graphs since we have **nine different side effects**

Network graph for nausea



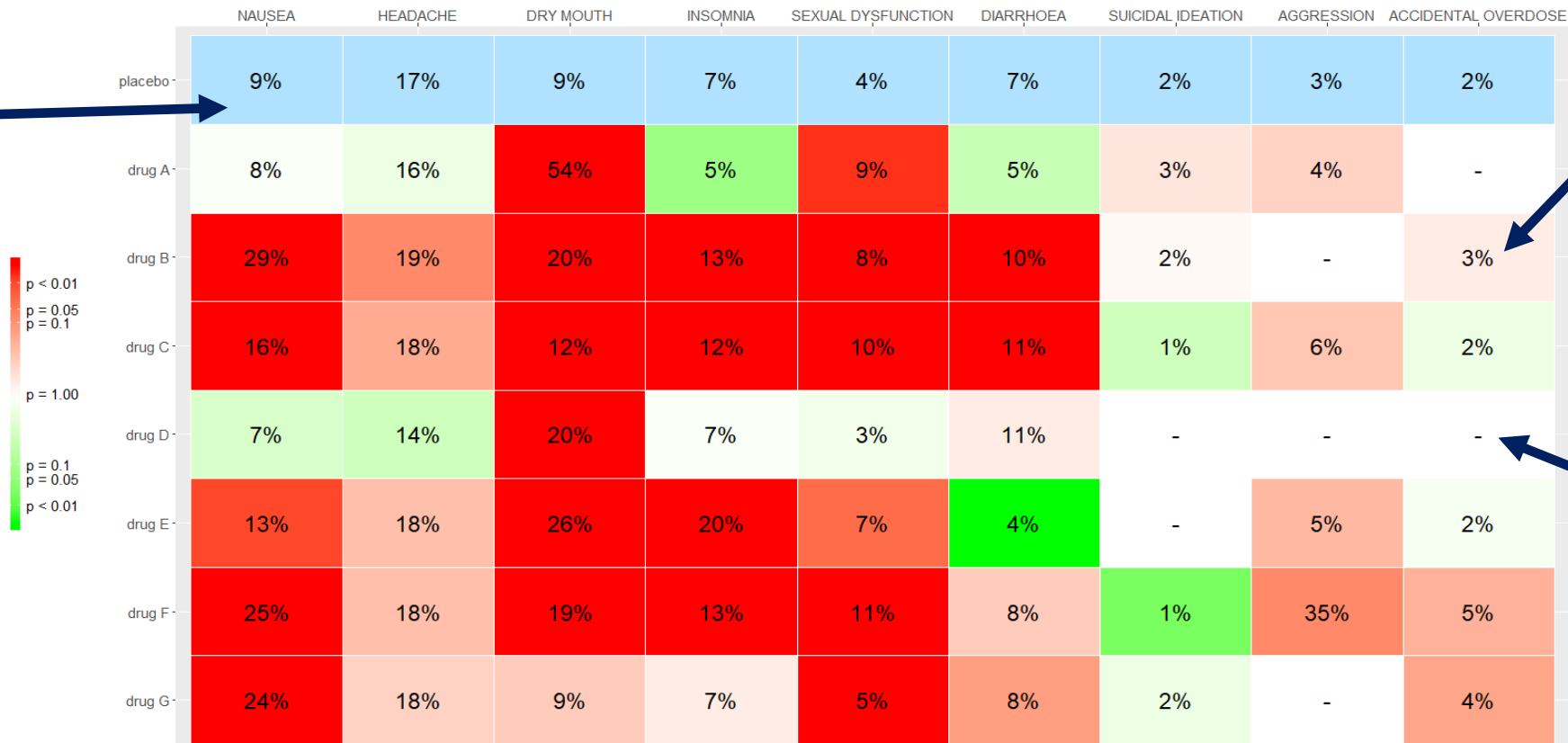
Drawing the Kilim plot

1. **Perform NMA** for all outcomes. This can be either a frequentist or a Bayesian NMA, using any NMA model.
2. **Perform a meta-analysis of the event rate** of the reference arms (or use external information) to obtain an estimate of the absolute event rate for each outcome, for the reference intervention.
3. **Combine the relative effects** from the first step and the **reference treatment event rate** from the second step in order to estimate the **absolute event rates for all treatments** in the network.
4. **Calculate Z-scores** using the estimated effect sizes and standard errors obtained from the NMAs of step one. For treatment X and outcome S , a Z-score equals to $Z_X^{(S)} = \log OR_X^{(S)} / SE(\log OR_X^{(S)})$
5. Use these information obtained (i.e. **Z-scores and absolute event rates**) to create the **Kilim plot!**

The Kilim plot

- We assume a negative (positive) Z-score is associated with a decrease (increase) in risk of side effects when using an active treatment as compared to placebo.
- **Green cells** denote a reduction in risk, **red cells** denote an increase in risk of a side effect

Estimates of absolute event rates are shown in the cells



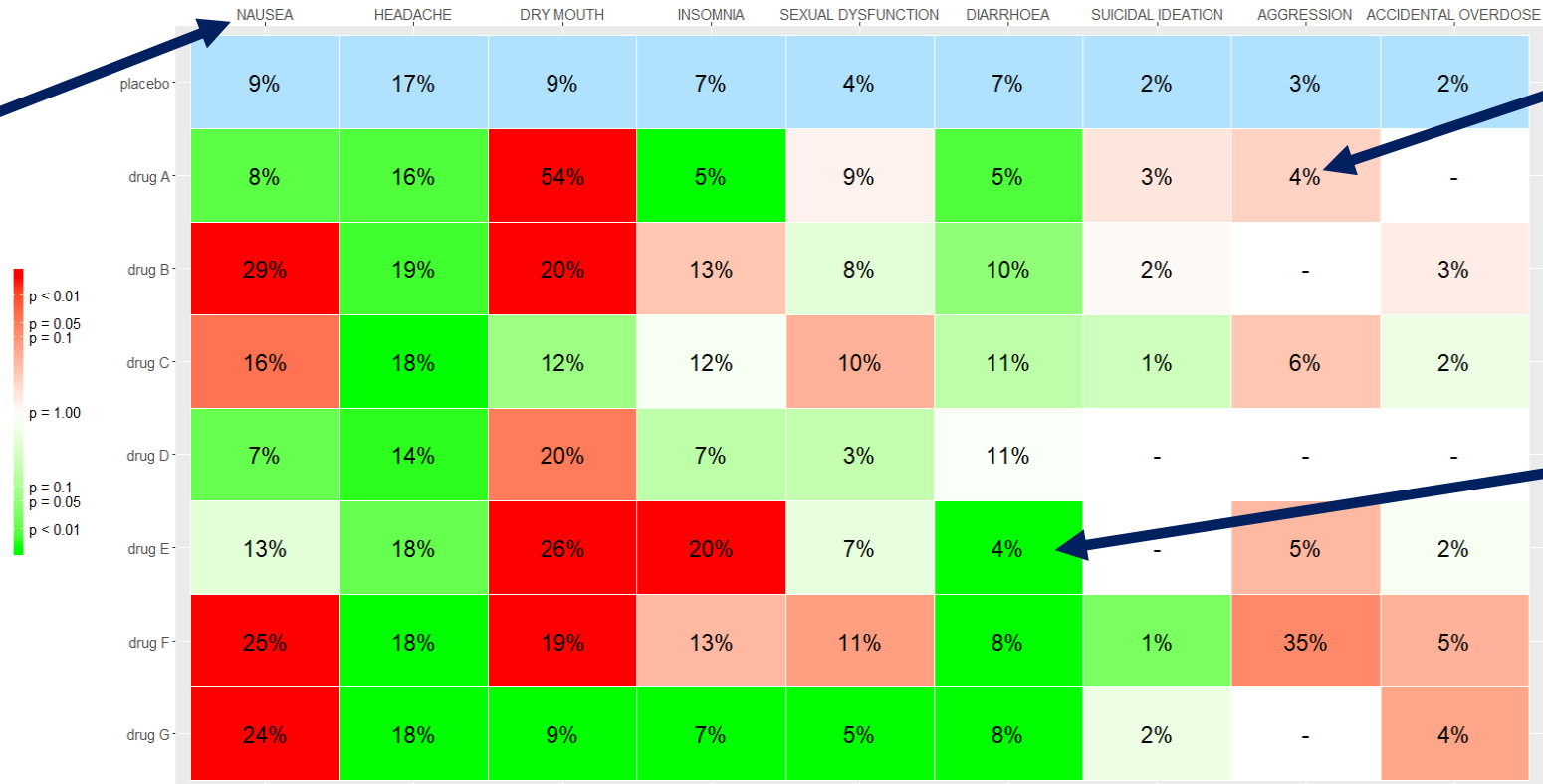
When confidence intervals of relative effects vs. placebo are very wide the corresponding colors tends to white

When a treatment is not included in the network the cell is empty

Incorporating clinically important values

- Assume a **clinically important value** $CIV_{RD}^{(S)}$ for the increase in risk compared to placebo, for outcome S . E.g. we deem that an increase of up to 5% in the risk of minor side effect S is acceptable.
- We use $CIV_{RD}^{(S)}$ to calculate $CIV_{OR}^{(S)}$, i.e. the clinically important value in the odds-ratio scale.
- We calculate $Z_X^{(S)} = (\log OR_X^{(S)} - CIV_{OR}^{(S)}) / SE(\log OR_X^{(S)})$, where X denotes a treatment, and use this to color the cells.

For non-serious outcomes (ie, all outcomes except suicidal ideation, aggression, accidental overdose) $CIV_{RD}^{(S)}$ is set to a risk increase of 5%. For serious outcomes we set $CIV_{RD}^{(S)} = 0$.



Estimated event rates remain the same, only colours change in cells according to the choice for $CIV_{RD}^{(S)}$.

Dark green cells for non-serious outcomes indicate **strong statistical evidence** that the corresponding drug leads to an increase of event rate **less than 5%** as compared to placebo, for the corresponding outcome.

Conclusion

- The **Kilim plot** can help in visualizing results from NMAs on multiple outcomes.
- It can be especially useful for **larger networks**, for the case of **many outcomes**, and when aiming to communicate NMA results with patients and/or clinicians, so as to facilitate **every-day clinical practice**.
- We illustrate the Kilim plot via an interactive web application (<https://esm.ispm.unibe.ch/shinies/kilim/>).
- Paper published in the Research Synthesis Methods Journal, <https://doi.org/10.1002/jrsm.1428>, R codes freely provided.

